

Seasonal Vegetation Analysis

The seasonal vegetation analysis was carried out over the Palakkad Area of Interest (AOI) using Sentinel-2 Surface Reflectance Harmonized imagery. The objective was to characterize seasonal vegetation behaviour and distinguish relatively persistent perennial vegetation, such as tree cover and agroforestry, from seasonally varying vegetation such as rice and other croplands.

1. Sentinel-2 Data Preparation

Sentinel-2 Surface Reflectance Harmonized imagery (COPERNICUS/S2_SR_HARMONIZED) was used. The imagery was filtered spatially using the Palakkad AOI and temporally according to two vegetation seasons:

- **Post-monsoon:** November 2024–February 2025
- **Pre-monsoon:** March–May 2025

Images with cloud cover greater than 60% were excluded. The QA60 band was used to mask cloud and cirrus-contaminated pixels. Sentinel-2 reflectance values were scaled using the standard factor of 0.0001. A median composite was generated for each season to reduce the influence of residual clouds and individual-image variations.

2. Spectral Vegetation Indices

Four spectral indices were derived for both seasonal composites:

- **NDVI (Normalized Difference Vegetation Index):** derived using the red (B4) and near-infrared (B8) bands to represent vegetation greenness and density.
- **EVI (Enhanced Vegetation Index):** derived using the blue (B2), red (B4), and near-infrared (B8) bands to provide additional sensitivity to vegetation and dense canopy conditions.
- **NDWI (Normalized Difference Water Index):** derived using green (B3) and near-infrared (B8) bands to support water and moisture discrimination.
- **NDMI (Normalized Difference Moisture Index):** derived using near-infrared (B8) and SWIR1 (B11) bands to characterize vegetation moisture conditions.

Thus, separate NDVI, EVI, NDWI and NDMI layers were produced for both the post-monsoon and pre-monsoon periods.

3. Seasonal Variability Analysis

To look at how plants grow and change with the seasons, the differences in NDVI and EVI were compared at the two periods. This was done by calculating the change in NDVI and EVI:

$$\Delta NDVI = NDVI_{pre-monsoon} - NDVI_{post-monsoon}$$

$$\Delta EVI = EVI_{pre-monsoon} - EVI_{post-monsoon}$$

The interpretation was based on the direction and magnitude of the change. Values close to zero indicate that vegetation response was similar between the two periods. Positive values indicate higher vegetation activity during the pre-monsoon period, while negative values indicate lower pre-monsoon activity relative to the post-monsoon period.

The absolute seasonal differences were also calculated:

$$NDVI \text{ Range} = |NDVI_{pre-monsoon} - NDVI_{post-monsoon}|$$

$$EVI \text{ Range} = |EVI_{pre-monsoon} - EVI_{post-monsoon}|$$

The areas where NDVI and EVI changed substantially are likely to represent places where seasonal crops such as rice undergo stronger vegetation changes. Areas where NDVI and EVI changed less and remained consistently high are likely to represent trees or other vegetation that remains relatively stable through the year.

4. Combined Seasonal Variability

NDVI and EVI variability were normalized and combined to produce a combined seasonal vegetation variability layer. This integrates the general vegetation response represented by NDVI with the additional canopy sensitivity provided by EVI.

The combined layer was interpreted as:

- **Low variability:** relatively stable vegetation response; potential perennial/tree-dominated vegetation.
- **High variability:** strong seasonal vegetation response; potential seasonal cropland.
- **Intermediate variability:** mixed or transitional vegetation conditions.

The seasonal analysis was used as a phenological assessment rather than a final land-cover classification. The resulting seasonal features were subsequently incorporated into the multi-season feature stack used for Random Forest classification.

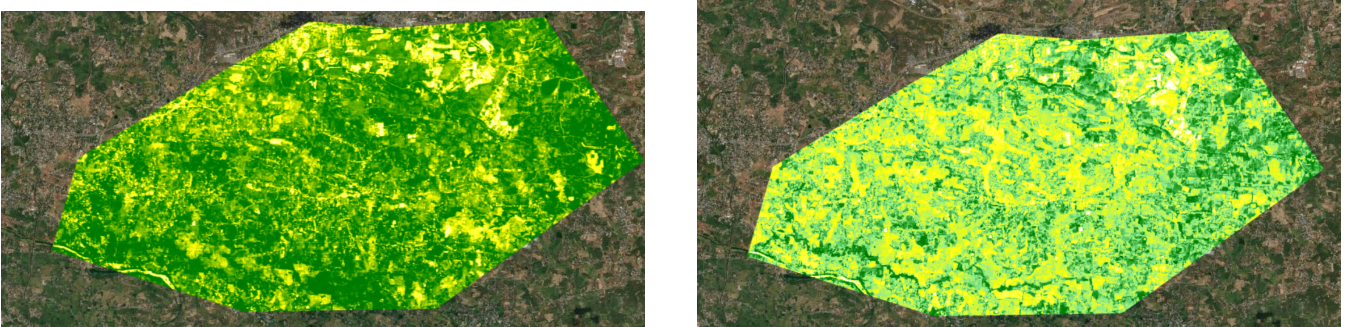


Figure 1: Seasonal vegetation post and pre monsoon index outputs for the Palakkad AOI.

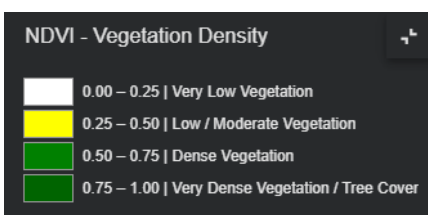


Figure 2: NDVI vegetation-density legend

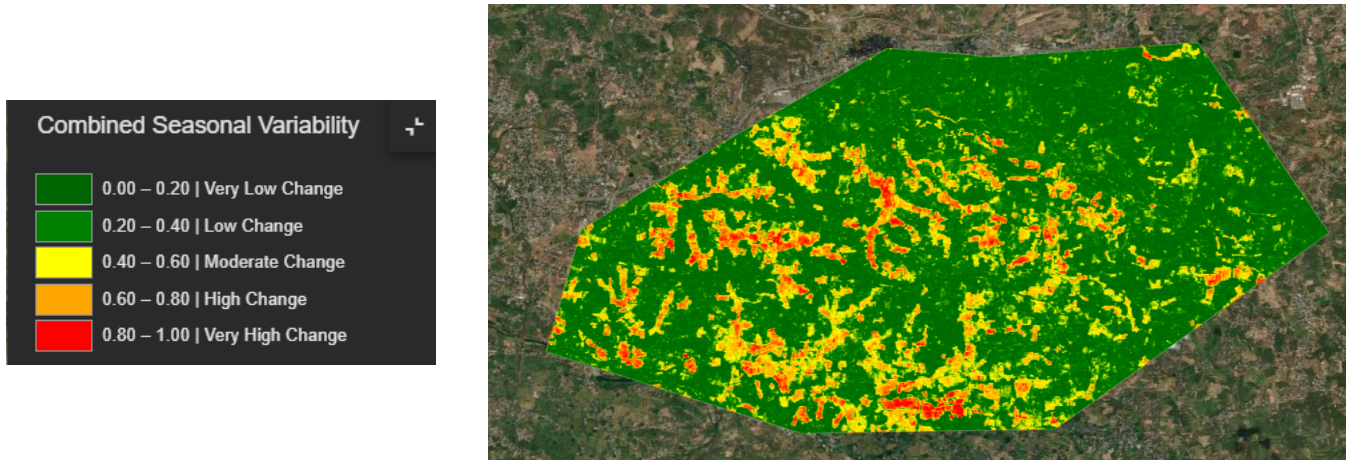


Figure 3: Combined seasonal vegetation variability legend and corresponding variability map.

5. Random Forest Classification and Training

- **Classifier:** Random Forest implemented in Google Earth Engine.
- **Land-cover classes:** Dense Forest, Agroforestry, Seasonal Cropland, Built-up, and Water.
- **Training data:** 351 manually digitized polygons interpreted using high-resolution satellite/basemap imagery.
- **Feature extraction:** Pixel-level features were extracted from a 22-band multi-season Sentinel-2 feature stack.
- **Features included:** NDVI, EVI, NDWI and NDMI; seasonal mean, standard deviation and range of NDVI and EVI; red-edge bands B5, B6, B7 and B8A; and post-monsoon and pre/summer observations.
- **Data split:** 70% for training and 30% for validation.
- **Model application:** The trained Random Forest was applied to the complete feature stack.
- **Output:** Final land-cover classification map of the Palakkad study area.

6. Classification Workflow



The resulting classification provides a spatial representation of Dense Forest, Agroforestry, Seasonal Cropland, Built-up and Water, while the independent validation dataset is used to assess the reliability of the classification using the confusion matrix, overall accuracy and Kappa coefficient.

7. Random Forest Classification and Accuracy Assessment

A Random Forest classifier was trained using the multi-season Sentinel-2 feature stack derived from manually digitized training polygons. The dataset contained 62,890 pixel samples from 346 polygons, divided at the polygon level into 47,537 training pixels from 259 polygons and 15,353 independent testing pixels from 87 polygons.

Five classes were mapped: Dense Forest, Agroforestry, Seasonal Cropland, Built-up, and Water.

The classifier achieved an overall accuracy of 94.8% and a Kappa coefficient of 0.875, indicating strong agreement between the classified and reference data.

Table 1: Class-wise precision, recall and F1-score.

Class	Precision	Recall	F1-score
Dense Forest	0.980	0.822	0.894
Agroforestry	0.813	0.842	0.827
Seasonal Cropland	0.972	0.974	0.973
Built-up	0.933	0.956	0.944
Water	0.795	0.769	0.782

The largest confusion occurred between Agroforestry and Seasonal Cropland, mainly because both can have similar vegetation responses and mixed pixels. Dense Forest was also partly confused with Agroforestry because both are tree-dominated classes.

Feature-importance analysis showed that summer red-edge bands (B5, B6, B7 and B8A), along with NDMI, NDWI, NDVI statistics and seasonal variability, were among the most important predictors. This demonstrates the value of combining spectral, vegetation, moisture and seasonal features.

A key limitation was the small number of Dense Forest training polygons (6), which may not represent the full variability of forest conditions. Additional reference polygons could improve classification performance. Overall, the 94.8% accuracy and 0.875 Kappa demonstrate that the multi-season Random Forest approach was effective for separating the five land-cover classes in the Palakkad study area.

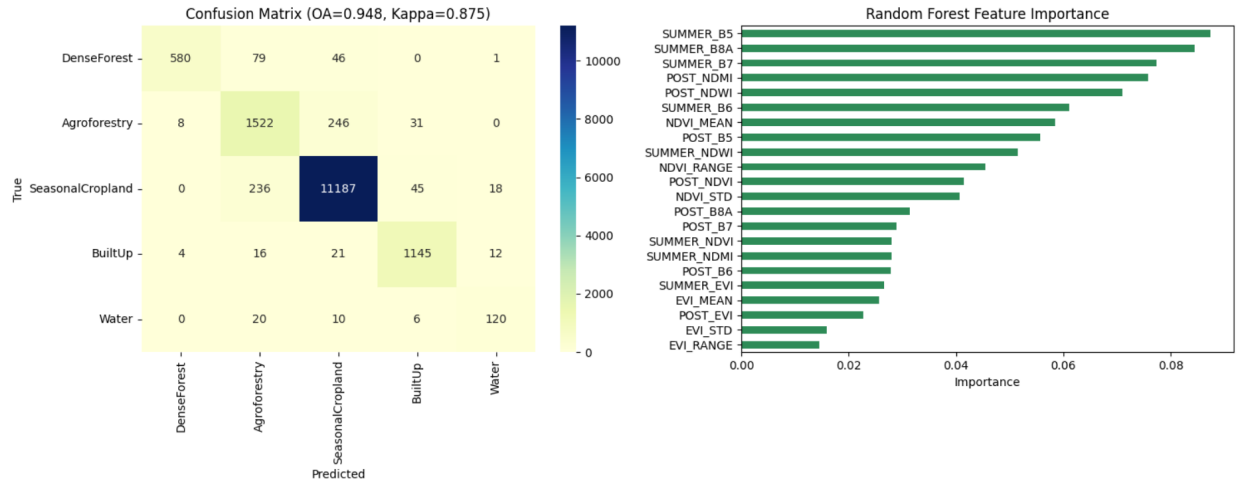


Figure 4: Confusion matrix and Random Forest feature-importance results.

8. Improvement Using SAR Data Fusion and Higher-Resolution Imagery

The classification could be further improved by integrating Synthetic Aperture Radar (SAR) data with the existing Sentinel-2 optical features. Sentinel-1 SAR observations, particularly VV and VH backscatter and their seasonal differences, can provide information on surface structure and moisture even when optical imagery is affected by cloud cover. Adding SAR features could help separate classes with similar optical vegetation responses, such as agroforestry and seasonal cropland, and could also improve the identification of water and structurally different vegetation.

Higher-resolution imagery could provide another improvement by reducing mixed-pixel effects and representing small land-cover features more clearly. Imagery with spatial resolution finer than Sentinel-2 could improve the delineation of small agricultural plots, narrow built-up areas, tree patches and boundaries between agroforestry and cropland. A practical approach would be to fuse the existing Sentinel-2 spectral and seasonal features with Sentinel-1 SAR variables and, where available, higher-resolution optical imagery. Additional field reference samples would further strengthen training and independent validation, particularly for the Dense Forest class where the current report identifies only six training polygons.

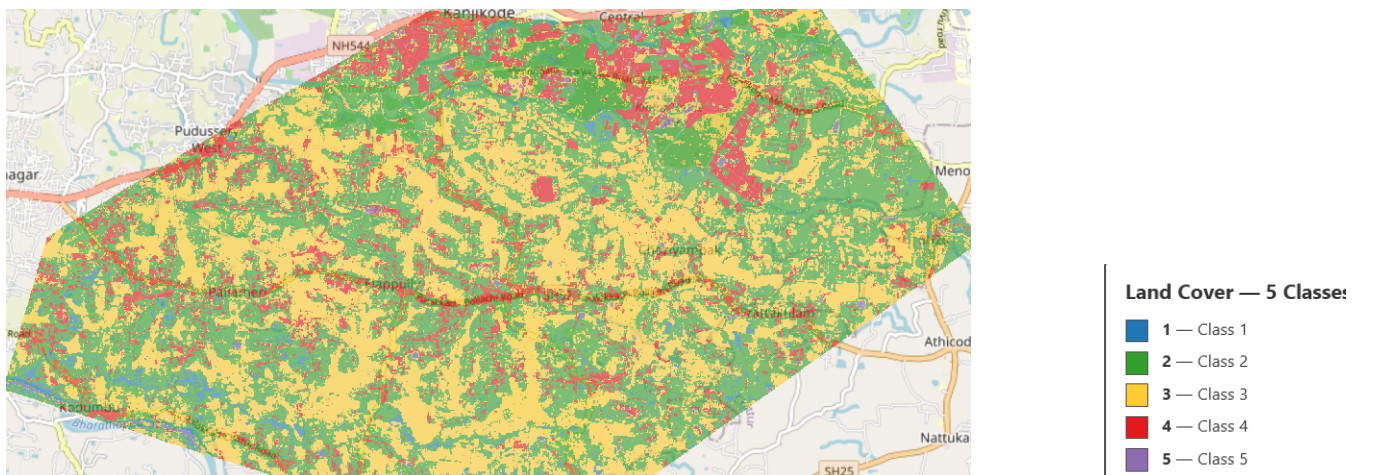


Figure 5: Final five-class land-cover classification map of the Palakkad study area and class legend.